HOTEL BOOKING MANAGEMENT SYSTEM USING ZOHO CREATOR

A PROJECT REPORT

Submitted by

KEERTHANA V

(241501082)

in partial fulfilment of the End Semester Examination for the course

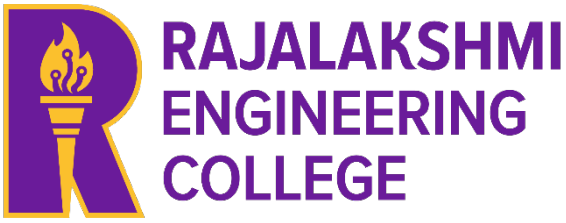
CB23F35 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF TECHNOLOGY

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Rajalakshmi Nagar, Thandalam, Tamil Nadu – 602 105

OCTOBER 2026

RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled "**Hotel Booking Management System using Zoho Creator**" is the bonafide work of **Keerthana V (241501082),** who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year **2026–2027**, **Semester V**, under my supervision.

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INTERNAL EXAMINER

EXTERNAL EXAMINER

DECLARATION

I, **Keerthana V (241501082),** hereby declare that the project report titled "**Hotel Booking Management System using Zoho Creator**", submitted for the End Semester Examination of the course **CB23F35 – Low-Code Application Development using Zoho Creator**, describes the Zoho Creator application built for this course – its data model, forms, reports, dashboard page, Deluge scripts and workflows.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. The first version of the application was generated with Zoho Creator's Zia AI application builder, and an AI assistant (Claude, by Anthropic) was used to configure and test parts of the application and to draft this report. These tools, and the defects and limitations found during testing, are acknowledged openly in Chapters 5, 6 and 8.

Place: Chennai
Date:

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I also acknowledge Zoho Corporation for providing access to the Zoho Creator platform and learning resources, and I thank my family and friends for their unwavering support.

KEERTHANA V

ABSTRACT

Small hotels often keep room details in a register, bookings in a notebook and totals in a spreadsheet. The same details are typed several times, an impossible stay (check-out before check-in) can be written down without any warning, bills are worked out by hand, and the manager has no single place to see which bookings are confirmed, checked in or cancelled.

This project builds a Hotel Booking Management System on the Zoho Creator low-code platform. The application has two forms, Room Inventory and Guest Booking, linked by a lookup field. The first version was generated with Zoho Creator's Zia AI builder and then reviewed and extended. A Deluge validation script blocks a booking whose check-out date is before the check-in date, a workflow sets a booking reference when the form loads, and another sends a confirmation email after a successful booking. Two list reports, two Kanban reports and a dashboard page give the manager a live view of rooms and bookings. Three permission sets, a Receptionist role, a room availability blueprint and a custom function, calculateStayCost, complete the application.

The application was tested on the live Zoho Creator app. A booking with check-out earlier than check-in was rejected with a clear error message, and a valid booking was saved and appeared in the report. Testing also found defects: the generated booking reference is malformed, and the total amount is not calculated because the custom function is not yet connected to the form. These are documented with their causes and proposed fixes. The work shows that a usable booking prototype can be produced in a short time with little code, and also that generated scripts must still be reviewed and tested.

Keywords: *Low-Code Development; Zoho Creator; Deluge Scripting; Zia AI Builder; Workflow Automation; Hotel Booking*

PROJECT AT A GLANCE

Attribute	Details
Project Title	Hotel Booking Management System using Zoho Creator
Application Domain	Hospitality / Hotel front-desk operations
Student Name & Register No.	Keerthana V – 241501082
Zoho Creator Edition	Zoho Creator trial (15-day trial plan)
Application Name (Link Name)	Hotel Booking Management (hotel-booking-management)
Published / Live URL	https://creatorapp.zoho.in/keerthana.v.2024.aiml_raj229/hotel-booking-management
Demo Credentials (per role)	Admin: account owner login Front desk: Receptionist role defined; no separate demo user created
Number of Forms / Subforms	2 / 0
Number of Reports (by type)	List: 2 Kanban: 2 Calendar: 0
Number of Pages / Dashboards	1 (Dashboard)
Number of Workflows	Form: 3 Schedule: 1 (disabled) Approval: 0 Blueprint: 1 Batch: 0
Deluge Functions / Custom Actions	1 function (calculateStayCost); no widget
Integrations	Email through the sendmail task; no external API
Roles & Profiles Defined	CEO (default) and Receptionist roles; 3 permission sets
Environments Used	None (application not published to Stage / Production)
Date of Production Deployment	Not deployed to Production (live application on trial plan, October 2026)

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
AI	Artificial Intelligence
API	Application Programming Interface
CO	Course Outcome
ER	Entity–Relationship
FK	Foreign Key
FR	Functional Requirement
HTML	HyperText Markup Language
LCAP	Low-Code Application Platform
NFR	Non-Functional Requirement
PK	Primary Key
RBAC	Role-Based Access Control
REST	Representational State Transfer
TC	Test Case
UAT	User Acceptance Testing
UC	Use Case
UI	User Interface
URL	Uniform Resource Locator
Zia	Zoho’s AI assistant

CHAPTER 1
INTRODUCTION

1.1 Background

Hotels of every size depend on a simple promise: when a guest arrives, the booked room is ready. Keeping that promise needs correct information about which rooms exist, what each room costs, which rooms are free, and which guest holds which room on which dates. In many small and mid-sized hotels this information still lives in a paper register, a telephone diary and a spreadsheet, and each of them is updated by a different person.

This project uses a realistic case: a small city hotel with a handful of rooms of different types (Single, Double and Suite) and a front desk that takes bookings by phone and at the counter. The aim is to move the whole booking process into one low-code application built on Zoho Creator, so that room data is stored once, bookings pick the room from a list, dates are checked automatically, and the manager can see occupancy and booking status on a live dashboard.

1.2 Problem Statement

The front desk records rooms, guests and bookings in separate places and relies on manual checks. This causes the following problems:

- Room details and prices are copied again and again into the register and the spreadsheet, so the two often disagree.
- A booking can be written with a check-out date earlier than the check-in date, and nobody notices until the guest arrives.
- The bill for a stay is calculated by hand from the number of nights and the room rate, which is slow and error-prone.
- The status of a booking (confirmed, checked in, checked out, cancelled) is not visible in one place.
- Guests do not receive a written confirmation, and there is no simple report for the manager.

1.3 Objectives

- To design a linked data model with two forms, Room Inventory and Guest Booking, connected by a lookup field.

- To validate booking dates automatically and prevent a check-out date that is earlier than the check-in date.
- To generate a booking reference and send a confirmation email when a booking is saved.
- To give the manager reports (list and Kanban views) and a dashboard showing totals and booking status counts.
- To define permission sets and a role for the hotel staff, and to study deployment, security and the limits of the low-code approach.

1.4 Scope of the Project

In scope:

- Room master data: room number, type, price per night, status, floor, amenities, occupancy, description and maintenance status.
- Guest booking entry with room lookup, stay dates, booking status, special requests and booking reference.
- Date validation, booking reference generation and confirmation email using Deluge workflows.
- List and Kanban reports, a dashboard page, permission sets, a role and a blueprint for room availability.
- Web access through the Zoho Creator live application.

Out of scope: online payment, channel-manager or booking-website integration, a guest self-service portal and a native mobile application. These are listed as future work in Chapter 9.

1.5 Justification for a Low-Code Approach

A hotel front desk needs forms, lists, a few business rules and a dashboard. These are exactly the building blocks that Zoho Creator provides, with hosting, login and permissions included. Compared with writing a web application from scratch, the low-code route is faster to build and easier for a non-programmer to maintain [1], [3], [6]. Zoho Creator also offers an AI builder, Zia, which can generate a first version of the application from a plain-language description [9]; this project uses that feature and then reviews, tests and extends the result.

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related low-code literature. Chapter 3 presents the requirements. Chapter 4 describes the design. Chapter 5 describes the implementation on Zoho Creator. Chapter 6 reports testing and debugging, including the defects found. Chapter 7 discusses deployment, performance, security and governance. Chapter 8 presents the results and a discussion of limitations, and Chapter 9 concludes with future work. The appendices contain the feature checklist, course outcome mapping, the Deluge source code, a user manual, the evaluation rubric and the viva-voce record sheet.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In a typical small hotel the receptionist takes a booking by telephone or at the counter, looks in a room register to see whether a room is free, writes the booking in a notebook and later types the details into an Excel sheet. The bill is worked out by hand at check-out, and the manager reads a weekly summary prepared from the sheet. Figure 2.1 shows this flow.

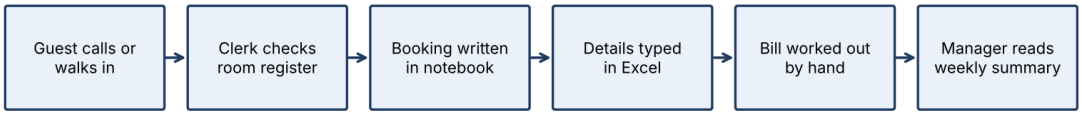


Figure 2.1: As-Is Process Flow of the Existing System

2.2 Limitations of the Existing System

- The same guest and room data is typed more than once, which wastes time and causes mismatches.
- There is no automatic date check, so impossible stays can be recorded.
- Room availability is not linked to bookings, so double booking is possible.
- Bills are calculated manually from nights and room rate.
- There is no confirmation message to the guest and no audit of who changed a booking.
- Reports are prepared by hand and are out of date as soon as they are made.

2.3 Low-Code versus Traditional Development

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code)	Low-Code on Zoho Creator
Development time	Several weeks for forms, database, API and UI	Days for forms, reports, dashboard and workflows
Skills required	Front-end, back-end, SQL and server set-up	Creator builder and basic Deluge scripting
Hosting and infrastructure	Own server or cloud account and backups	Hosted and backed up by Zoho

Criterion	Traditional (Pro-Code)	Low-Code on Zoho Creator
Security and access control	Login, roles and encryption coded by the team	Built-in permission sets, roles and data sharing
Change management	Code change, testing and redeployment	Edit a field or rule in the builder
Flexibility	Unlimited	Limited to platform features and extension points
Cost	Developer and hosting cost	Subscription per user (15-day trial used here)

2.4 Literature Survey

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	Richardson & Rymer, Forrester, 2014 [3]	New Development Platforms Emerge for Customer-Facing Applications	Introduced the term low-code for platforms that speed up delivery of business applications.	Defines the approach used here.
2	Sahay et al., 2020 [5]	Supporting the understanding and comparison of low-code development platforms	Compares platforms on data modelling, workflow, integration and deployment features.	Basis for judging Zoho Creator's features.
3	Bock & Frank, 2021 [6]	Low-Code Platform	Describes typical parts of a low-code platform and its limits.	Used in the limitations discussion (Section 8.4).
4	Di Ruscio et al., 2022 [7]	Low-code development and model-driven engineering	Relates low-code to model-driven engineering and discusses open challenges.	Supports the data-model-first design.
5	Zoho Corporation [1], [2], [9]	Zoho Creator, Deluge and Zia documentation	Official reference for forms, workflows, functions and AI-assisted building.	Main technical reference for implementation.

2.5 Emerging Trends in Low-Code Development

Three trends shaped this work. First, AI-assisted building: Zoho Creator's Zia can create a complete application, including forms, reports, a dashboard, permissions, a blueprint and sample data, from a text prompt [9], which is how the first version of this project was produced. Second, citizen development, where business users configure applications with little code, supported by governance features such as roles and environments. Third, hybrid pro-code extension, where custom functions or external services handle logic that the visual tools cannot express [1], [2], [6].

2.6 Proposed System

The proposed system is a Zoho Creator application called Hotel Booking Management. It removes the limitations of Section 2.2 as follows:

- Room data is entered once in Room Inventory; bookings pick the room through a lookup field.
- A Deluge validation script blocks a booking whose check-out date is before the check-in date.
- A workflow generates a booking reference and another sends a confirmation email on successful submission.
- List and Kanban reports and a dashboard always show current data.
- Permission sets and a role limit what hotel staff can do.
- A custom function, calculateStayCost, holds the cost rule (nights multiplied by price per night) for reuse.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Administrator / Hotel Manager	Owner of the application	Configure the app, manage rooms, view dashboard and reports, manage roles and permissions.
Front-Desk Staff (Receptionist)	Handles guests at the counter	Create bookings, update booking status at check-in and check-out.
Housekeeping Supervisor	Looks after room readiness	Update cleaning and maintenance status of rooms.
Guest	Customer who books a room	Provides details; receives the confirmation email (no login in this version).

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Status
FR-01	The system shall store room details: number, type, price per night, status, floor, amenities, occupancy, description and maintenance status.	Admin	Implemented
FR-02	The system shall record a guest booking with name, email, phone, room, check-in and check-out dates, status and special requests.	Staff	Implemented
FR-03	The room shall be chosen from the Room Inventory list (lookup).	Staff	Implemented
FR-04	The system shall reject a booking whose check-out date is before the check-in date.	System	Implemented and tested
FR-05	The system shall generate a booking reference when the form loads.	System	Implemented; format defect (D1)
FR-06	The system shall send a confirmation email to the guest after a successful booking.	System	Implemented; delivery not verified
FR-07	The system shall calculate the total amount as nights multiplied by price per night.	System	Function created; not connected to the form (D2)
FR-08	The system shall show booking status counts, total amount and room counts on a dashboard.	Admin	Implemented
FR-09	The system shall provide list and Kanban reports of bookings and rooms.	Admin, Staff	Implemented
FR-10	The system shall limit actions by permission set and role.	Admin	Partly implemented; not tested with users

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Forms and reports should open within a few seconds on a normal connection.
NFR-02	Security	Only signed-in users can open the application; actions are limited by permission sets.
NFR-03	Usability	Fields have clear labels; drop-downs and lookups reduce typing.
NFR-04	Availability	The application is hosted by Zoho and available whenever the platform is.
NFR-05	Maintainability	Rules are short, commented Deluge scripts, one per purpose.
NFR-06	Auditability	Zoho Creator records added time and modified time for each record.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator (15-day trial plan), data centre zoho.in
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment)
Browser	Google Chrome (latest version)
API testing tool	Not used in this version
Other Zoho services	Email through the Creator sendmail task
Client hardware	Laptop or desktop with an internet connection

3.5 Use Case Model

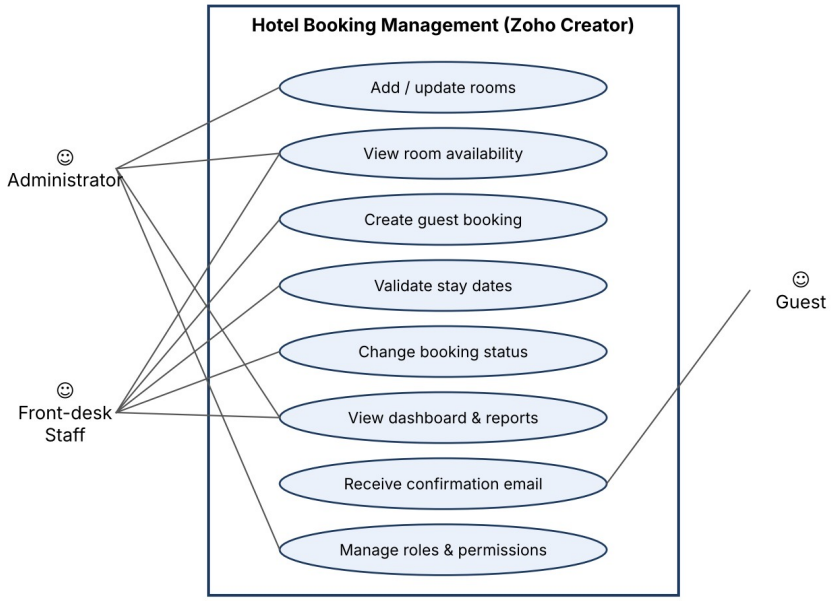


Figure 3.1: Use Case Diagram of the Hotel Booking Management System

Table 3.5: Use Case Description - Create a Guest Booking

Item	Details
Use Case ID	UC-01
Use Case Name	Create a Guest Booking
Actor(s)	Front-Desk Staff
Pre-condition	Rooms exist in Room Inventory.
Main Flow	1. Staff opens the Guest Booking form. 2. The system sets a booking reference. 3. Staff enters guest details, selects the room and enters check-in and check-out dates. 4. Staff clicks Submit. 5. The system validates the dates, saves the record and sends the confirmation email.
Alternate Flow	4a. Check-out date is before check-in date: the system shows an error message and does not save.
Post-condition	The booking appears in the Guest Booking Report and in the dashboard counts.

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

The application runs entirely on the Zoho Creator platform. Users open the live application in a browser. Forms, reports and the dashboard page are served by Creator, data is stored in the Creator data store, and Deluge workflows run on form events. The confirmation email is sent through Creator's sendmail task. The first version of the application was generated by the Zia AI builder (Figure 4.1).

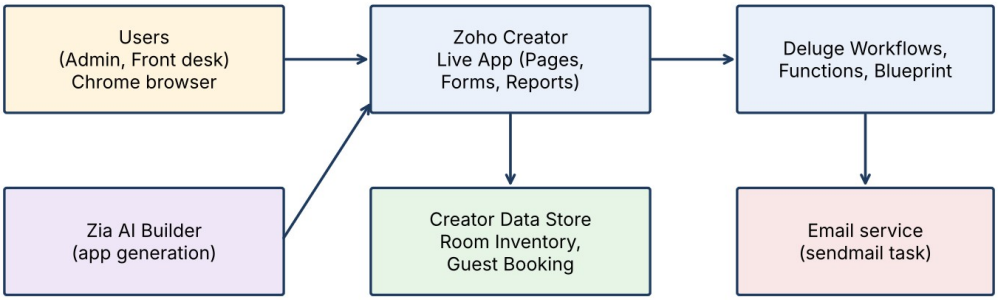


Figure 4.1: System Architecture of the Hotel Booking Management System

4.2 Data Model

The data model has two entities. One room can have many bookings, and each booking refers to exactly one room through a lookup field (Figure 4.2, Table 4.1). Zia proposed the same model when the application was generated (Figure 5.1).

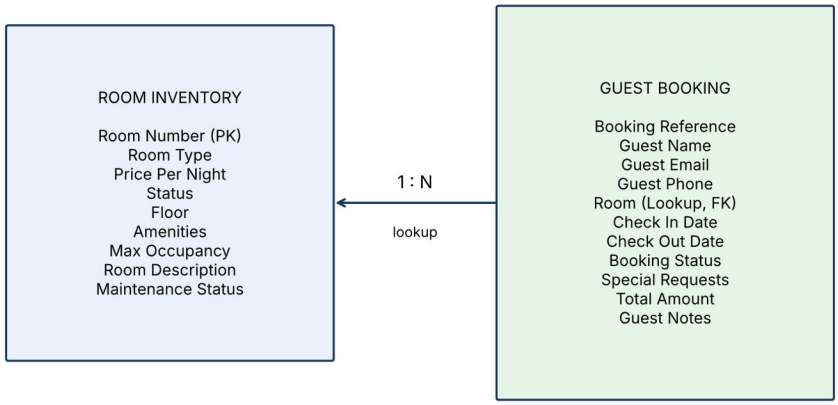


Figure 4.2: Entity-Relationship Diagram of the Hotel Booking Management System

Table 4.1: Entities and Relationships

Entity	Description	Relationship
Room Inventory	Master list of rooms with type, price, status and amenities	One room is referred to by many bookings
Guest Booking	One record for each reservation	Room field is a lookup to Room Inventory

4.3 Form Design

Field types below are taken from how each field appears in the form builder and live form.

Table 4.2: Field Specification - Guest Booking Form

Field Label	Field Type	Notes
Guest Name	Name	First name shown in the form
Guest Email	Email	Used as the recipient of the confirmation email
Guest Phone	Phone	Country code selector (India +91 by default)
Room	Lookup	Pulls the room number from Room Inventory
Check In Date	Date	Format dd-MMM-yyyy
Check Out Date	Date	Checked by the validation script
Booking Status	Drop Down	CheckedIn, Confirmed, Cancelled, CheckedOut
Special Requests	Multi Line	Free text
Total Amount	Currency	Intended to hold nights x price (see defect D2)
Booking Reference	Single Line	Set on form load by a workflow
Guest Notes	Multi Line	Free text

Table 4.3: Field Specification - Room Inventory Form

Field Label	Field Type	Notes
Room Number	Single Line	Unique room number, for example 101 to 105
Room Type	Drop Down	Single, Double, Suite; colour-coded in the report
Price Per Night	Currency	Dollar currency as generated by Zia
Status	Drop Down	Available or Occupied
Floor	Number	Floor of the room
Amenities	Multi Select	TV, Wi-Fi, Balcony, MiniBar, PoolAccess
Max Occupancy	Number	Maximum guests
Room Description	Single Line	For example Comfort Queen Room
Maintenance Status	Drop Down	Operational or UnderMaintenance

4.4 Process and Workflow Design

Figure 4.3 shows the booking process. When the form loads a booking reference is set. When the user submits, the validation workflow compares the two dates. If the check-out date is earlier the submit is cancelled with an error message; otherwise the record is saved and the confirmation email workflow runs.

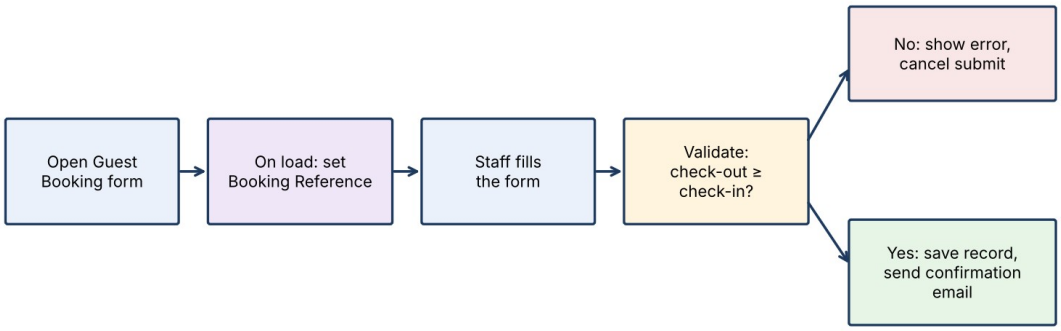


Figure 4.3: Workflow / Process Flow for Creating a Booking

4.5 User Interface and Navigation Design

The application menu has three entries: Dashboard, Guest Booking and Room Inventories. Each of the last two groups a form with its reports (Figure 4.4).

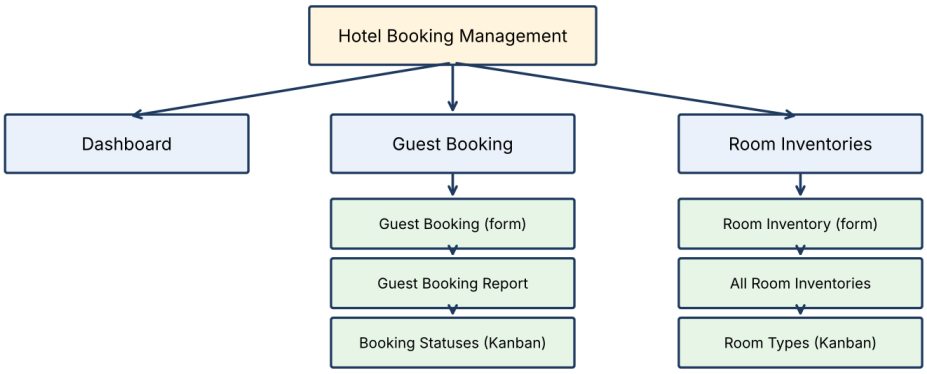


Figure 4.4: Navigation Structure of the Hotel Booking Management System

4.6 Access Control Design

Zia created three permission sets, described in Table 4.4, and the Receptionist role was added under the default CEO role.

Table 4.4: Permission Sets and Intended Users

Permission Set	Description (as generated)	Intended User
Front Desk Agent	Manage guest check-ins and check-outs by updating booking statuses	Receptionist

Permission Set	Description (as generated)	Intended User
Housekeeping Supervisor	Update room cleaning records and maintenance statuses	Housekeeping
Inventory Administrator	Create and maintain the master list of rooms including types and amenities	Manager

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The application was created in the author's Zoho Creator trial account with the Create using Zia option. A plain-language description of a hotel booking system (room inventory, guest bookings, automatic price calculation and reservation status tracking) was given to Zia, which proposed use cases and a data model and then generated the application (Figures 5.1 and 5.2). The generated application was then reviewed and extended by hand as described in the following sections.

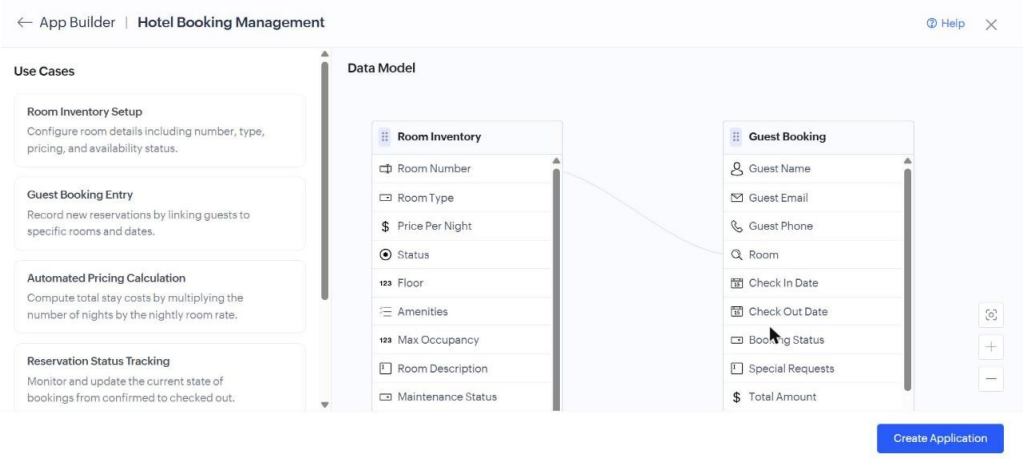


Figure 5.1: Zia Proposed Use Cases and Data Model

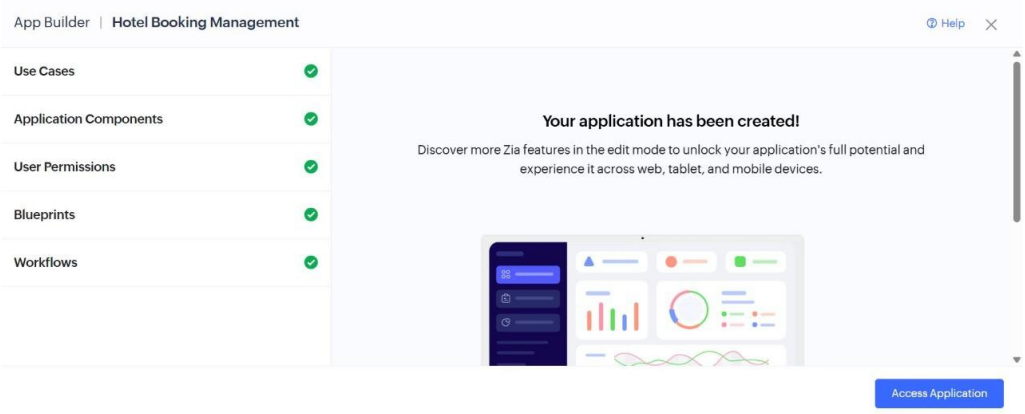


Figure 5.2: Zia Confirmation That the Application Has Been Created

5.2 Forms and Data Capture

Two forms were generated: Room Inventory and Guest Booking. Figure 5.3 shows the Guest Booking form in the form builder with the field properties panel, and Figures 5.4 and 5.5 show the live form.

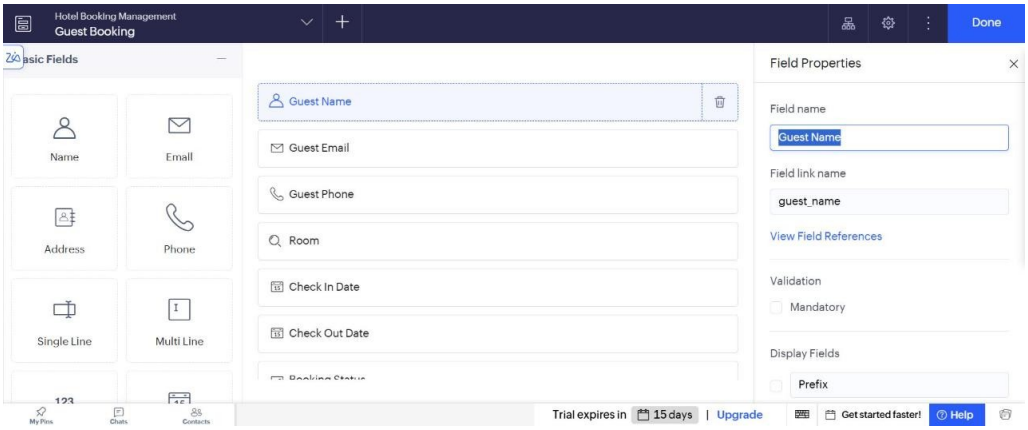


Figure 5.3: Guest Booking Form in the Form Builder

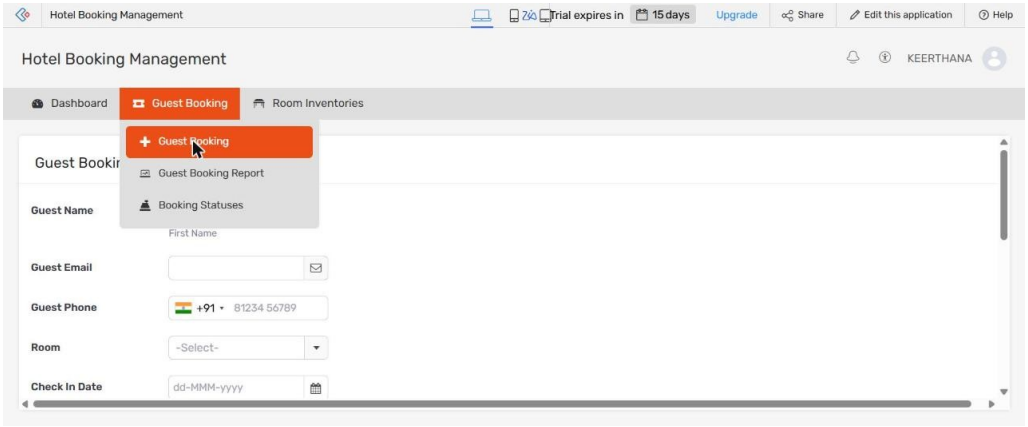


Figure 5.4: Guest Booking Form in Live Mode (Top Section)

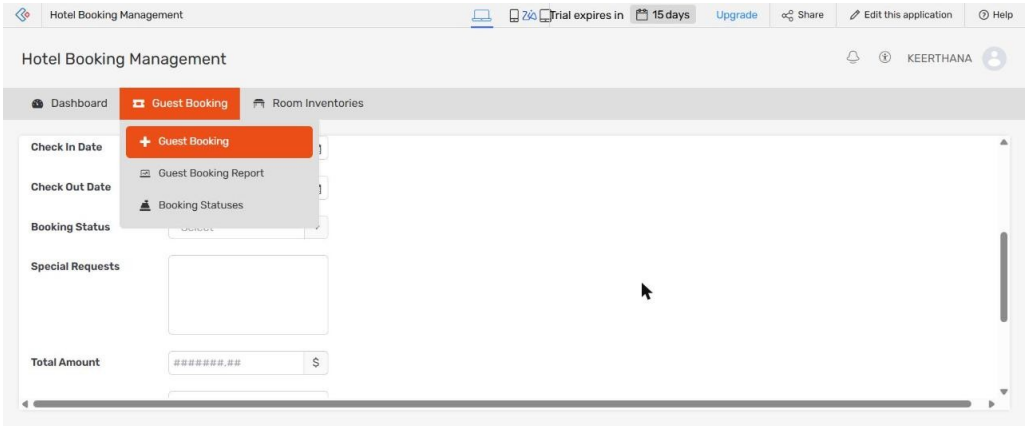


Figure 5.5: Guest Booking Form in Live Mode (Lower Section)

5.3 Reports

Table 5.1: Reports Implemented

Report	Type	Based on	Purpose
Guest Booking Report	List	Guest Booking	All bookings with guest, room, dates, status and requests
All Room Inventories	List	Room Inventory	All rooms with type, price, status, floor and amenities
Booking Statuses	Kanban	Guest Booking	Bookings grouped by status

Report	Type	Based on	Purpose
Room Types	Kanban	Room Inventory	Rooms grouped by type

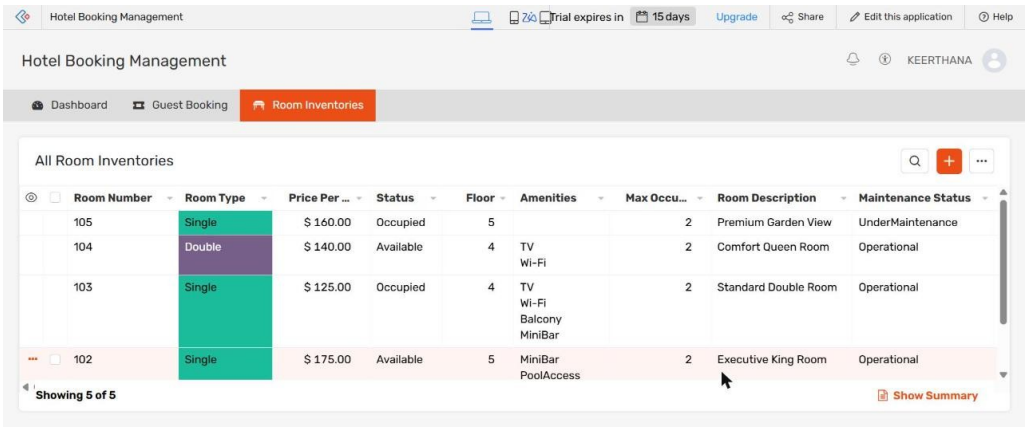


Figure 5.6: All Room Inventories Report

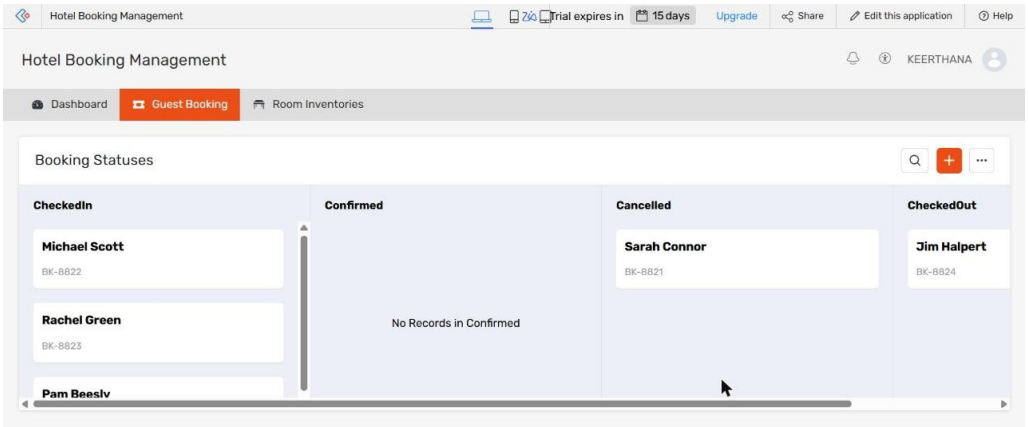


Figure 5.7: Booking Statuses Kanban Report

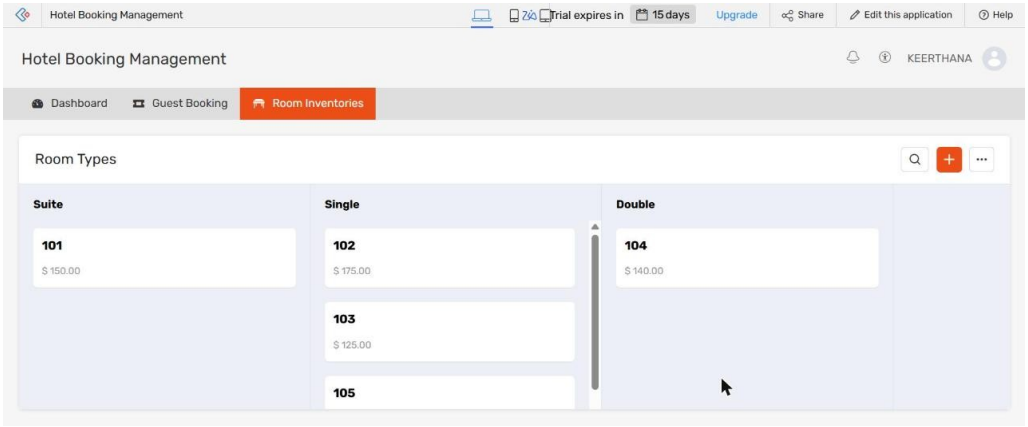


Figure 5.8: Room Types Kanban Report

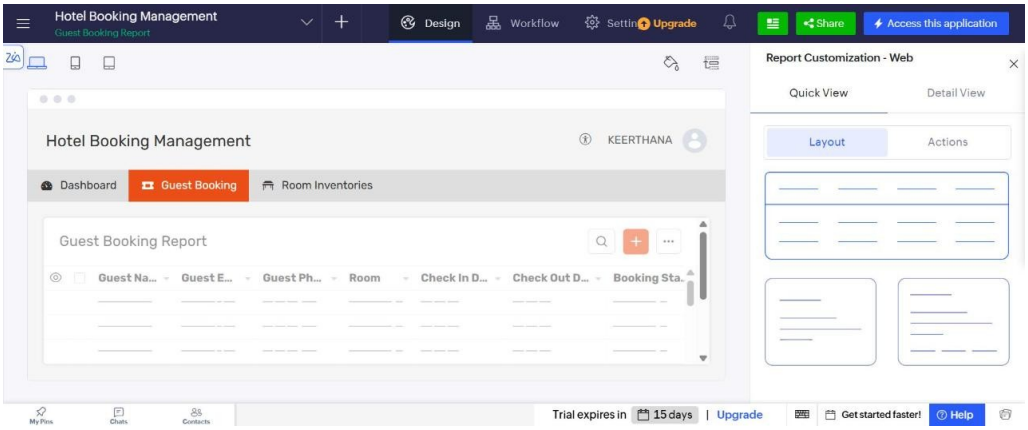


Figure 5.9: Report Customization Panel in the Builder

5.4 Pages and Dashboards

The home page of the application is a dashboard page generated by Zia. It shows four counters (Total Amount, Total Price Per Night, Count of Guest Booking and Total Room Inventories), four booking status counters and quick links to add or view records. Figure 5.10 shows the live page and Figure 5.11 shows the same page in the page builder. No additional custom page was designed manually.

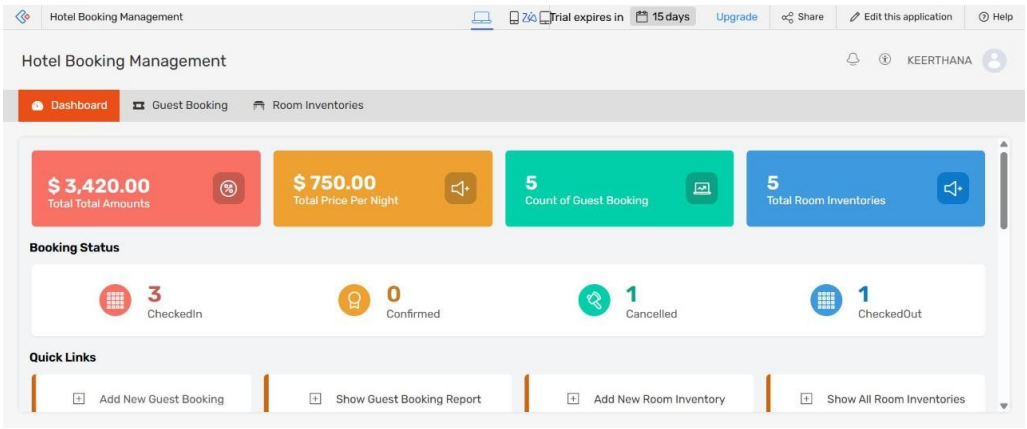


Figure 5.10: Dashboard Page in Live Mode

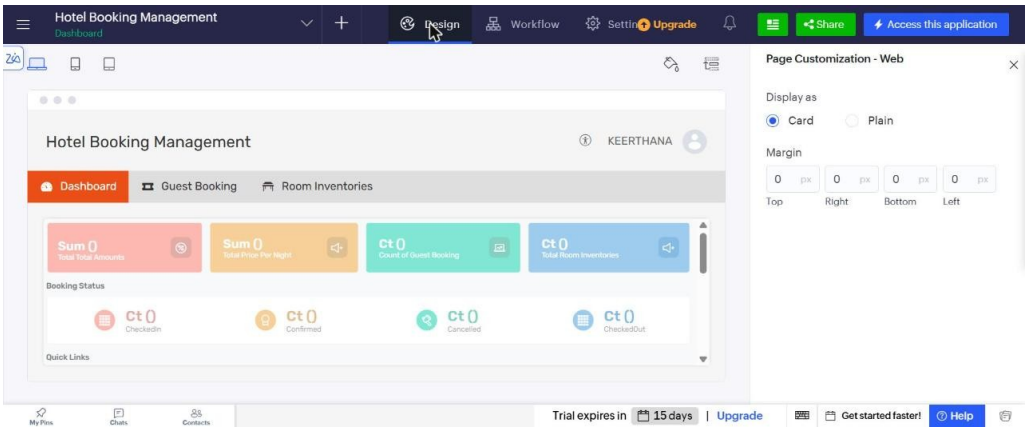


Figure 5.11: Dashboard Page in the Page Builder

5.5 Deluge Scripting

Table 5.2: Deluge Scripts Implemented

Script	Event	Purpose
Auto-Set Booking Reference	Guest Booking - On load	Builds a booking reference from the date, time and user ID
Validate Check-out Date	Guest Booking - Validation on submission	Cancels submit if check-out is before check-in
Send Confirmation Email	Guest Booking - On successful submission	Builds an HTML message and sends it to the guest email
calculateStayCost	Custom function	Returns nights multiplied by price per night

The validation script, which was tested, is shown below. The complete source of all scripts is in Appendix C.

```
if(input.check_out_date != null && input.check_in_date != null)
{
    if(input.check_out_date < input.check_in_date)
    {
        alert "Error: Check-out date cannot be before check-in date.";
        cancel submit;
    }
}
```

The first three scripts were generated by Zia and reviewed by the author. The function calculateStayCost was created by hand in the Functions section of the workflow builder (Figure 5.16).

5.6 Workflows and Automation

Table 5.3: Workflows Implemented

Workflow	Type	Trigger	Status
Auto-Set Booking Reference	Form	On load of Guest Booking	Enabled
Send Confirmation Email	Form	On successful submission	Enabled
Validate Check-out Date	Form	Validation on submission	Enabled
Room Availability Update	Blueprint	Room Inventory	Enabled
Daily Maintenance Alert	Schedule	Date-time schedule	Disabled

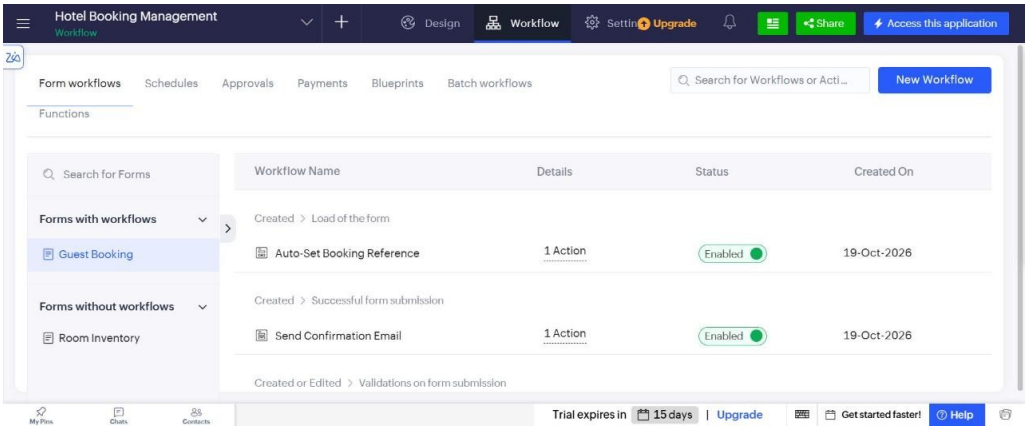


Figure 5.12: Form Workflows of Guest Booking

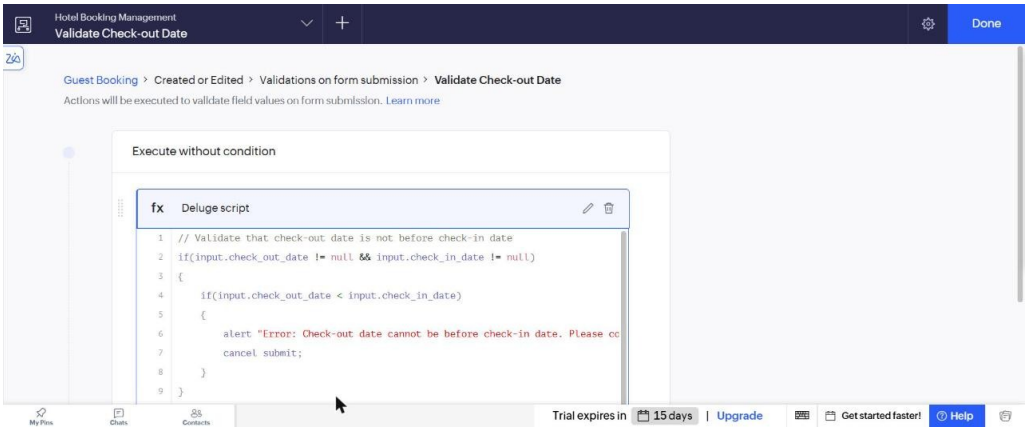


Figure 5.13: Validate Check-out Date Workflow with Deluge Script

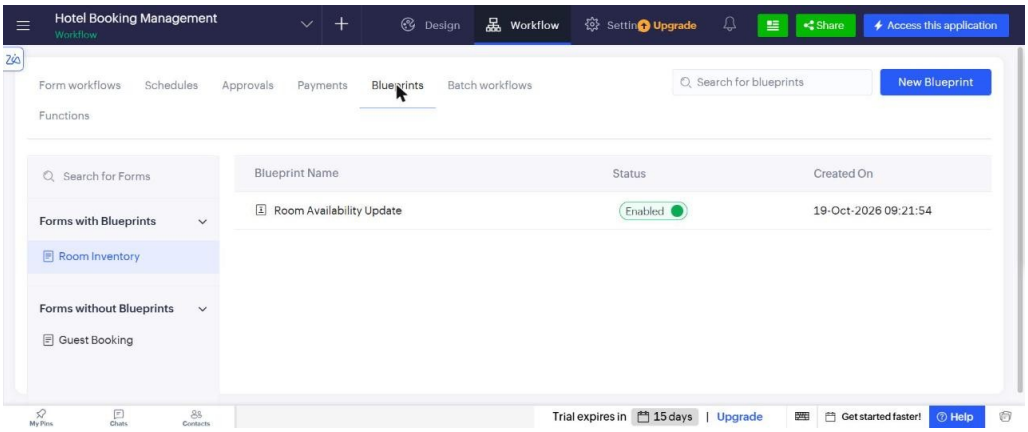


Figure 5.14: Room Availability Update Blueprint

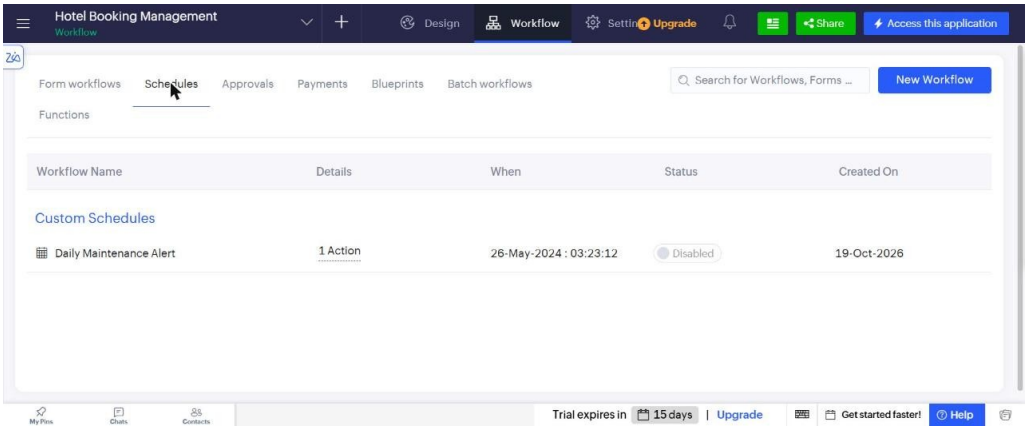


Figure 5.15: Schedule Workflow (Daily Maintenance Alert, Disabled)

No approval workflow or batch workflow was created for this project.

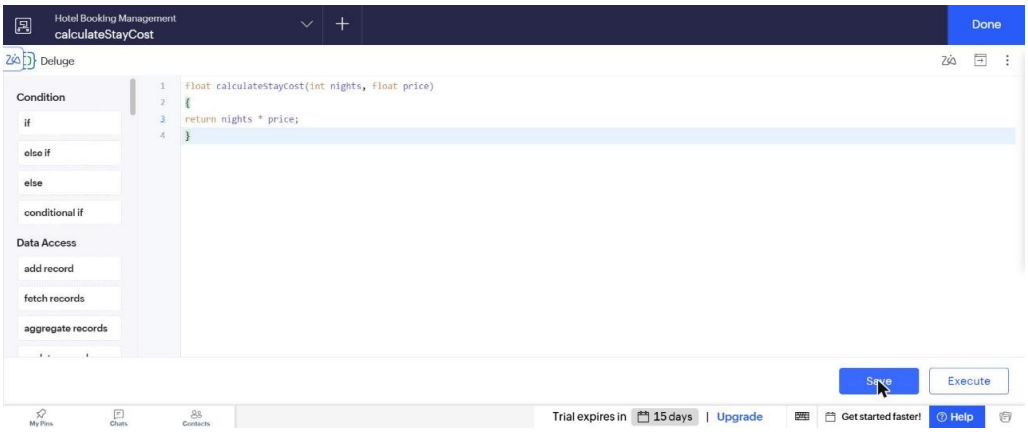


Figure 5.16: Custom Function calculateStayCost in the Editor

5.7 Integrations

Table 5.4: Integrations Implemented

Integration	Mechanism	Used for
Email	Creator sendmail task (sender: logged-in user)	Booking confirmation to the guest
Other Zoho services / REST APIs	Not configured	Listed as future work

No Zoho Connection and no external API call was created, and no REST API test tool was used in this version of the project.

5.8 User Management and Role-Based Access

Figures 5.17 and 5.18 show the permission sets generated by Zia and the role hierarchy after the Receptionist role was added. No test users were invited, so the permission sets were not exercised by a second login.

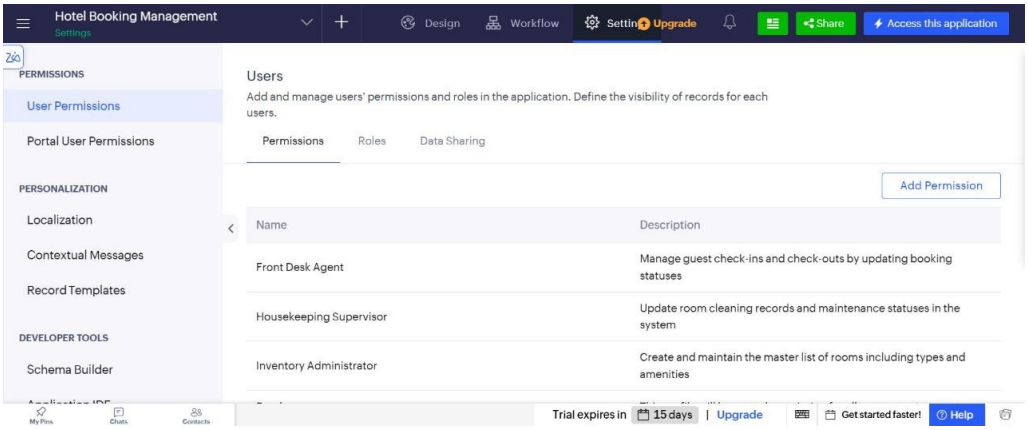


Figure 5.17: Permission Sets

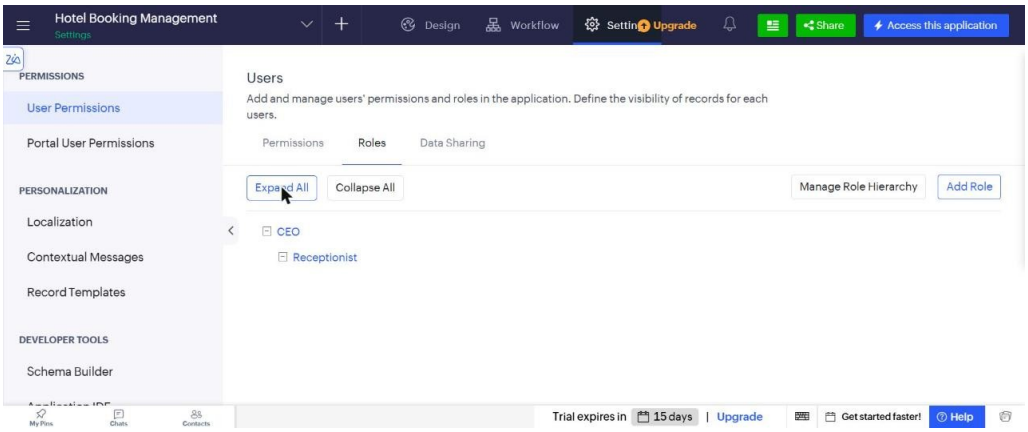


Figure 5.18: Role Hierarchy with the Receptionist Role

5.9 AI and Analytics Features

The AI feature used is the Zia build agent, which produced the data model, forms, reports, dashboard, permission sets, blueprint, workflows and sample data (five rooms and five bookings). The built-in analytics are the dashboard counters and the report summary option. Other Zia features, such as prediction or natural-language analytics, were not explored.

5.10 Pro-Code Extension

Pro-code extension in this project is limited to the custom Deluge function calculateStayCost(int nights, float price), which returns nights x price. It is saved but not yet called from the booking form, so Total Amount is not filled automatically (defect D2). No Zoho Catalyst function or widget was built.

5.11 Mobile and Portal Access

The page builder shows desktop, tablet and phone previews of the generated pages. A guest portal and the Zoho Creator mobile application were not configured or tested.

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing was done manually on the live application in Google Chrome. Each test case gives the input, the expected result and the actual result seen on screen. Where a feature could not be checked, the result is stated as not verified rather than passed. A test booking with dummy guest details (guest email on the reserved domain example.com) was used so that no real person received an email.

6.2 Test Cases

Table 6.1: Test Cases

TC ID	Test Case	Expected Result	Actual Result	Status
TC-01	Submit booking with check-in 06-Oct-2026 and check-out 01-Oct-2026	Error message; booking not saved	Dialog: Error: Check-out date cannot be before check-in date. Please correct the dates and try again. Record not saved (Figure 6.1)	Pass
TC-02	Submit booking with check-in 06-Oct-2026 and check-out 09-Oct-2026	Booking saved	Data Added Successfully; record visible in the report (Figures 6.2, 6.3)	Pass
TC-03	Booking reference set on form load	Readable reference	A value is set, but its format is wrong (defect D1)	Fail
TC-04	Total Amount calculated from nights and price	Amount filled automatically	Total Amount stayed blank	Fail
TC-05	Confirmation email after submission	Email sent to guest	Workflow ran without an error message; delivery to the test address was not verified	Not verified
TC-06	Guest Booking Report shows seeded records	Five sample bookings with status colours	Five records shown with colours (Figure 5.4 and report)	Pass
TC-07	Dashboard counters	Totals and status counts shown	Total amount $3,420.00; 5 bookings; 5 rooms; CheckedIn 3, Confirmed 0, Cancelled 1, CheckedOut 1 (Figure 5.10)	Pass
TC-08	Kanban reports	Cards grouped by status and room type	Both Kanban views opened (Figures 5.7, 5.8)	Pass
TC-09	Access with a second user role	Limited access	Not tested (no second user invited)	Not tested

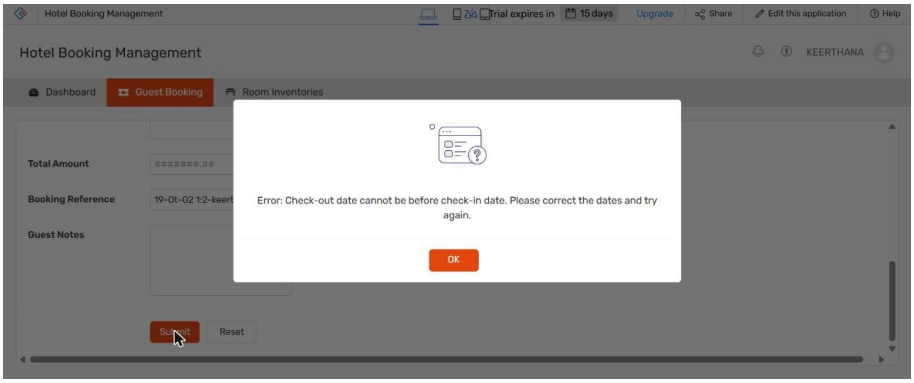


Figure 6.1: TC-01: Error Message for Check-out Before Check-in

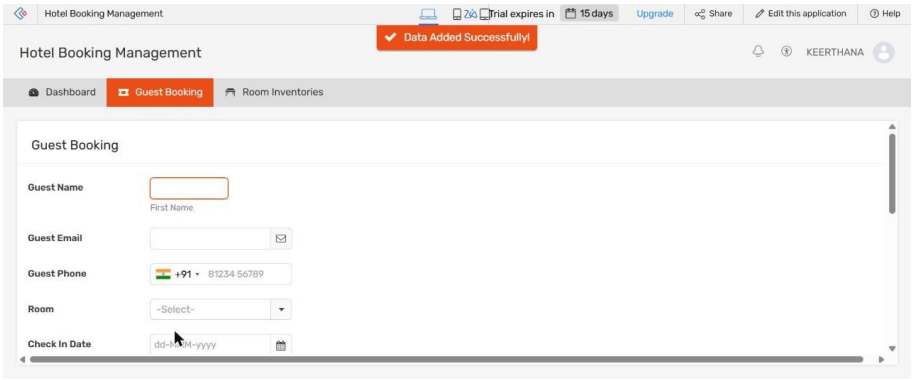


Figure 6.2: TC-02: Data Added Successfully

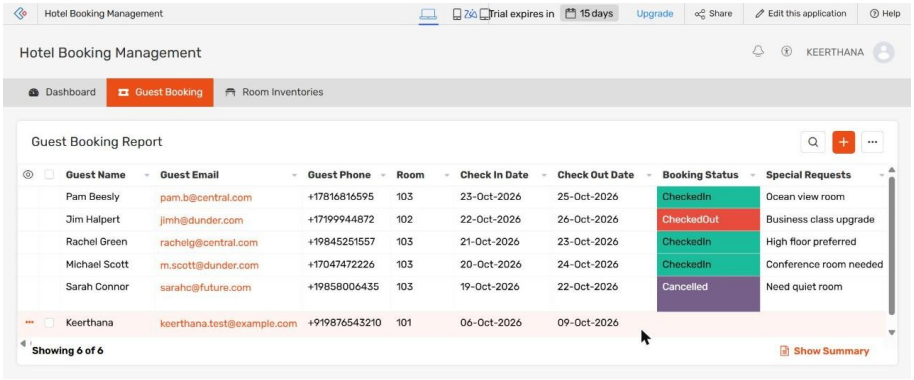


Figure 6.3: TC-02: New Booking Visible in the Guest Booking Report

6.3 Debugging Deluge Scripts

The booking reference seen in the form during testing was 19-Ot-02 1:2-keerthana.v.2024... instead of a value such as 202610191200-userid. Reading the script showed the cause. The workflow takes zoho.currentdate and cuts it with substring(0,4), substring(5,7) and so on, which assumes the text is in the form 2026-10-19. In this account the date is returned as 19-Oct-2026, so the cuts pick the wrong characters. A suggested fix is to format the values explicitly:

```
bookingRef = zoho.currentdate.toString("yyyyMMdd") +
zoho.currenttime.toString("HHmm") + "-" + zoho.loginuserid;
```

This correction is proposed but was not applied to the submitted application. The same test showed that Total Amount stays blank, because the function calculateStayCost is not connected to the form. The info and alert statements and the Deluge execution log are the tools used for such debugging.

6.4 Defect Log

Table 6.2: Defect Log

ID	Description	Severity	Status
D1	Booking reference is built with wrong substring positions and shows garbled characters	Medium	Open; fix proposed in Section 6.3
D2	Total Amount is not calculated; calculateStayCost is not wired to the form	High	Open
D3	Daily Maintenance Alert schedule is disabled and carries a past date (26-May-2024)	Low	Open
D4	Booking Status is not mandatory; a booking was saved without a status	Low	Open
D5	Email delivery could not be confirmed with the dummy address	Low	Open

6.5 User Acceptance Testing

User acceptance testing with hotel staff was not carried out for this project. Table 6.3 is a self-assessment against the objectives.

Table 6.3: Acceptance Self-Assessment

Objective	Self-Assessment
Linked data model and forms	Met
Date validation	Met (tested)
Booking reference and confirmation email	Partly met (D1, D5)
Automatic amount calculation	Not met (D2)
Reports and dashboard	Met
Roles and permissions	Partly met (not tested with users)

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

Zoho Creator provides Development, Stage and Production environments so that changes can be tested before end users see them. The hotel application was built in the account's default environment and is used as a live application on the trial plan. It was not published through Stage and Production, so there is no separate production version.

Table 7.1: Deployment Checklist

Check	Status
Application builds and opens in live mode	Done
Core rule (date validation) tested	Done
Known defects fixed	Not done (D1 to D5)
Test users and role testing	Not done
Published to Stage and Production	Not done
Live URL	https://creatorapp.zoho.in/ keerthana.v.2024.aiml_raj229/hotel-booking-management

7.2 Performance Optimisation

No measured performance tuning was done, because the application holds only a handful of records. Table 7.2 lists the design choices that support performance and the techniques that would matter as data grows.

Table 7.2: Performance Optimisation Considerations

Technique	Applied?	Comment
Lookup instead of copying room data	Yes	Room details are stored once in Room Inventory
Report pagination	Yes (default)	Reports show records in pages
Criteria and limits when fetching records in Deluge	Not needed yet	Current scripts do not fetch records
Avoid fetch inside loops	Not needed yet	No loops in current scripts
SubForms for repeated data	No	Not required by this data model

7.3 Security Features

Table 7.3: Security Controls

Control	Status
Sign-in required for the application	Provided by Zoho login

Control	Status
Permission sets (three)	Created by Zia
Role hierarchy with Receptionist role	Configured
Input validation on dates	Implemented
Data sharing rules between roles	Not configured
Testing with a second user	Not done

7.4 Governance and Administration

The account owner is the administrator and can edit the application, manage roles and publish changes. For a real deployment the hotel should name an owner for the application, review permission sets before onboarding staff, keep a test environment for changes, and record changes to scripts in a change log.

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The application opens on the dashboard (Figure 5.10), lets staff create bookings (Figures 5.4 and 5.5), rejects an invalid stay (Figure 6.1), saves a valid one (Figure 6.2) and shows it in the report (Figure 6.3). The Kanban reports give the manager a view by status and by room type (Figures 5.7 and 5.8).

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective	Outcome	Evidence
Linked data model with two forms	Achieved	Figures 4.2, 5.3; Tables 4.2, 4.3
Automatic date validation	Achieved and tested	Figure 6.1; TC-01
Booking reference and confirmation email	Partly achieved	TC-03, TC-05; defects D1, D5
Reports and dashboard	Achieved	Figures 5.6 to 5.11
Permission sets, role, deployment study	Partly achieved	Figures 5.17, 5.18; Chapter 7

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Aspect	Existing System	Proposed Application
Room data	Register and spreadsheet	Single Room Inventory form
Date check	None	Automatic validation
Booking status view	Manual	Kanban and list reports
Guest confirmation	None	Email workflow (delivery not verified)
Bill calculation	By hand	Function created, not yet connected
Manager summary	Weekly, by hand	Live dashboard

8.4 Limitations of the Low-Code Approach

- The generated scripts still need review: the booking reference script contains a bug that the platform did not catch.
- Visual tools hide logic, so a wrong assumption (such as a date format) is found only by testing.
- Platform limits on records, API calls and script run time matter for a large hotel chain with many properties and a channel manager.

- The application depends on one vendor's platform, and moving it elsewhere would mean rebuilding it.
- Highly custom user interfaces or offline use need pro-code extensions.

8.5 Reflection on the No-Code, Low-Code and Pro-Code Spectrum

This project sits mostly at the no-code end: Zia produced the application from a prompt, and forms, reports and the dashboard needed no code. The workflows are low-code, because they are short Deluge scripts that had to be read and tested. The custom function is a small step towards pro-code. A fuller project would move further along the spectrum by wiring the function into the form and calling an external service, as discussed in Section 9.2.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

A working hotel booking prototype was produced on Zoho Creator in a short time. It has linked Room Inventory and Guest Booking forms, date validation that was tested and works, list and Kanban reports, a dashboard, permission sets and a custom function. Testing also found real defects: the booking reference is malformed and the total amount is not calculated. These are recorded openly in Chapter 6. The work shows that low-code and AI-assisted building can deliver a useful first version quickly, and that testing and code review remain necessary.

9.2 Future Enhancements

- Fix defect D1 with explicit date formatting and call calculateStayCost from the form to fill Total Amount (D2).
- Make Booking Status mandatory with a default of Confirmed, and block double booking of the same room for overlapping dates.
- Update Room Inventory status automatically when a booking is checked in or out.
- Add an approval workflow for cancellations and a schedule for check-in reminders.
- Publish through Stage and Production environments and test with real staff users.
- Add a guest portal, online payment and integration with Zoho CRM or Zoho Books.

9.3 Learning Outcomes and Career Relevance

The project gave practice in data modelling, form design, Deluge scripting, workflows, roles and permissions, testing and debugging, and in judging the output of an AI builder critically. These skills apply to roles such as low-code developer, business automation specialist and platform administrator, and they combine well with core programming skills.

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APPENDIX A - ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1: Feature Coverage Checklist

Feature	Covered	Remarks
Forms with lookup field	Yes	Guest Booking to Room Inventory
Multi-select, drop-down, date, email, phone fields	Yes	Tables 4.2, 4.3
SubForm	No	Not needed by the data model
Formula, auto number, QR or signature fields	No	Not used
List report	Yes	Two list reports
Kanban report	Yes	Two Kanban reports
Calendar report	No	Not created
Page / dashboard (UI Builder)	Partly	Generated by Zia, not hand-designed
Form workflow (Deluge)	Yes	Three form workflows
Schedule workflow	Partly	Created, disabled
Approval workflow	No	Not created
Blueprint	Yes	Room Availability Update
Batch workflow	No	Not created
Custom function	Yes	calculateStayCost (not wired)
Email integration	Partly	Sendmail used; delivery not verified
Zoho connection / REST API	No	Not configured
Roles and permission sets	Partly	Created; not tested with users
Zia AI builder	Yes	Application generated
Environments (Stage / Production)	No	Not used
Mobile / portal	No	Not configured

APPENDIX B - COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2	Good	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1-5.6), 6	Good	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7-5.9	Satisfactory	
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7	Needs Improvement	
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3	Good	

APPENDIX C - COMPLETE DELUGE SOURCE CODE

The scripts below were transcribed from the workflow editor of the live application. Long lines are shown as seen on screen; lines not visible during transcription are marked with an ellipsis. The application itself is the authoritative source.

C.1 Auto-Set Booking Reference (Guest Booking, On load)

```
// Generate unique booking reference using current date, time, and user ID
if(isBlank(input.booking_reference))
{
    // Get current date and time components
    currentDate = zoho.currentdate;
    currentTime = zoho.currenttime;
    // Extract year, month, day, hour, minute from current date/time
    year = currentDate.substring(0,4);
    month = currentDate.substring(5,7);
    day = currentDate.substring(8,10);
    hour = currentTime.substring(11,13);
    minute = currentTime.substring(14,16);
    // Get logged in user ID for uniqueness
    userId = zoho.loginuserid;
    // Generate unique booking reference: YYYYMMDDHHMM-USERID
    bookingRef =
year.concat(month).concat(day).concat(hour).concat(minute).concat(...);
    // Set the booking reference value
    input.booking_reference = bookingRef;
}
```

C.2 Validate Check-out Date (Guest Booking, Validation on submission)

```
// Validate that check-out date is not before check-in date
if(input.check_out_date != null && input.check_in_date != null)
{
    if(input.check_out_date < input.check_in_date)
    {
        alert "Error: Check-out date cannot be before check-in date. Please
correct the dates and try again.";
        cancel submit;
    }
}
```

C.3 Send Confirmation Email (Guest Booking, On successful submission)

```
// Fetch guest email from input
if(input.guest_email.isNull())
{
    info "Guest email is missing";
}
else
{
    // Prepare booking details for email
    bookingReference = input.booking_reference;
    guestName = input.guest_name.first_name.concat(" ").concat(...);
    checkInDate = input.check_in_date.toString("dd-MMM-yyyy");
    checkOutDate = input.check_out_date.toString("dd-MMM-yyyy");
    roomNumber = input.room.room_number;
    totalAmount = input.total_amount.toString();
    // Construct email message
    emailSubject = "Booking Confirmation - " + bookingReference;
    emailMessage = "<p>Dear " + guestName + ",</p>";
    emailMessage = emailMessage.concat("<p>Your booking has been
successfully ...");
    ...
    emailMessage = emailMessage.concat("<p><strong>Check-out Date:</strong>
" ...);
```

```
    emailMessage = emailMessage.concat("<p><strong>Total Amount:</strong> "
+ ...);
    emailMessage = emailMessage.concat("<p>Thank you for choosing our hotel!
</...");
    // Send confirmation email
    sendmail
    [
        from :zoho.loginuserid
        to :input.guest_email
        subject :emailSubject
        ...
    ]
}
```

C.4 calculateStayCost (Custom function)

```
float calculateStayCost(int nights, float price)
{
    return nights * price;
}
```

APPENDIX D - USER MANUAL

D.1 Accessing the Application

Open https://creatorapp.zoho.in/keerthana.v.2024.aiml_raj229/hotel-booking-management in Google Chrome and sign in with the Zoho account that owns or has been invited to the application. The menu shows Dashboard, Guest Booking and Room Inventories.

D.2 Instructions for the Hotel Manager (Administrator)

- Open Dashboard to see total amount, booking counts and booking status counts.
- Open Room Inventories, then Room Inventory to add a room, or All Room Inventories to view and edit rooms.
- Use Room Types and Booking Statuses to see rooms and bookings in Kanban view.
- Use Edit this application to change forms, workflows, roles and permissions.

D.3 Instructions for Front-Desk Staff

- Open Guest Booking, then Guest Booking to add a booking.
- Enter guest name, email and phone, choose the room, and enter check-in and check-out dates. Choose a Booking Status.
- Click Submit. If the check-out date is before the check-in date, correct the dates and submit again.
- Open Guest Booking Report to find a booking and update its status at check-in and check-out.

APPENDIX E - EVALUATION RUBRIC

Student Name	Keerthana V	Register No.	241501082
Project Title	Hotel Booking Management System	Date	

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		100	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90-100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70-89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50-69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner	Signature of External Examiner
Name: _____	Name: _____

APPENDIX F - VIVA-VOCE RECORD SHEET

Student Name	Keerthana V	Register No.	241501082
Department / Section	Artificial Intelligence and Machine Learning	Date of Viva	
Application URL Verified		Demo Duration	_____ min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E - Excellent, G - Good, S - Satisfactory, NI - Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1-8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	100	

Examiners' Remarks:

_____

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_____

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INTERNAL EXAMINER	EXTERNAL EXAMINER
Name: _____	Name: _____